ST2334 Cheat Sheet, AY26/27 S1

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1 Basic Concepts of Probability

1.1 Probability

Probability of an event A is a measure of the likelihood of the event occurring.

Relative frequency of A, when experiment is repeated n times and event A occurs m times, approaches $P(A)$ as $n \to \infty$:

$$P(A) = \lim_{\{n\to\infty\}} \frac{m}{n}$$

$$0 \le P(A) \le 1$$

$$P(S) = 1$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

1.2 Conditional Probability

Conditional Probability ($P(A \mid B)$) is probability of A occurring given that B occurred.

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B|A)}{P(B)}$$

Simpson's Paradox

Phenomenon where a trend appears in different groups of data but disappears or reverses when the groups are combined. Or formally,

$$P(A \mid B \cap C_i) \ge P(A \mid B' \cap C_i) \forall i \text{ but } P(A \mid B) < P(A \mid B')$$

Multiplication Rule

$$P(A \cap B) = P(A \mid B)P(B) \text{ if } P(B) > 0$$

1.3 Independence

Events A and B are **independent**, $A \perp B \Leftrightarrow$

$$P(A \cap B) = P(A)P(B)$$

Tautologies:

- Suppose P(A) > 0 and P(B) > 0, If $A \perp B$, then A and B are not mutually exclusive (aka $A \cap B \neq \varnothing$)
- $S \perp A$ for all events A; $\varnothing \perp A$ for all events A
- If $A \perp B$, then $A \perp B'$, $A' \perp B$, $A' \perp B'$
- Independence cannot be expressed in terms of venn diagram

1.4 Bayes' Theorem

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B \mid A)P(A)}{\sum_{\{i=1\}}^{n} P(B \mid A_i)P(A_i)}$$

Law of Total Probability

Suppose $A_1, A_2, ..., A_n$ is a partition of the sample space S

$$P(B) = \sum_{\{i=1\}}^{n} P(B \cap A_i) = \sum_{\{i=1\}}^{n} P(B \mid A_i)P(A_i)$$

$$P(B) = P(A)P(B \mid A) + P(A')P(B \mid A')$$

2 Random Variable

Random Variable is a function that transform a sample from sample space to a real number.

$$X : S \to \mathbb{R} : s \in S \text{ and } X(s) = x \in \mathbb{R}$$

Range space of X is set of real numbers that satisfies

$$R_X = \{x \mid X(s) = x \forall s \in S\} : R_X \in \mathbb{R}$$

Subsets of Sample Space

$$\{X = x\} = \{s \in S : X(s) = x\}, \{X = x\} \in S$$
$$\{X \in A\} = \{s \in S : X(s) \in A\}, \{X \in A\} \in S$$

2.1 Probability Distribution

Probability Mass Function (pmf)

Discrete random variable X_D, when $R_X = \{x_1, x_2, ...\}$ is finite or countable.

pmf, $f(x)$ for $\{X = x\}$

$$f(x) = \begin{cases} P(X = x) \mid \forall x \in R_X \\ 0 \mid \forall x \notin R_X \end{cases}$$

Well-defined pmf for discrete random variable X:

1. $\forall x_i \in R_X, f(x_i) \ge 0$
2. $\forall x_i \notin R_X, f(x_i) = 0$
3. $\sum_{i=1}^{\infty} f(x_i) = \sum_{x_i \in R_X} f(x_i) = 1$

Probability Density Function (pdf)

Continuous random variable X_C, when $R_X \in [a, b]$ is an interval(s).

pdf, $P(a \le X \le b)$ for any a and b such that $a \le b$

$$P(a \le X \le b) = \int_a^b f(x)dx$$

Well-defined pdf for continuous random variable X:

1. $\forall x \in R_X, f(x) \ge 0$
2. $\forall x \notin R_X, f(x) = 0$
3. $\int_{R_X} f(x)dx = 1$

Other key properties,

- Specific point is zero: $P(X = x_i) = \int_{x_i}^{x_i} f(x)dx = 0$
- Endpoint doesn't matter: $P(a \le X \le b) = P(a < X < b)$

2.2 Cumulative Distribution Function

Cumulative Distribution Function (cdf), F(x) for any random variable X is defined as

$$F(x) = P(X \le x) = \begin{cases} f(0) + f(1) + ... + f(x) \text{ if X is discrete} \\ \int_{-\infty}^{x} f(t)dt \text{ if X is continuous} \end{cases}$$

$$P(a \le X \le b) = P(X \le b) - P(X < a) = F(b) - F(a-)$$
$$P(X = x) = P(x \le X \le x) = F(x) - F(x-)$$

Let $F(x) = \int_{-\infty}^{x} f(t)dt$

$$f(x) = F'(x) = \frac{d}{dx}\int_{-\infty}^{x} f(t)dt$$

2.3 Expectation and Variance

Expectations or mean of X is defined by: (μ_X may not be in R_X)

$$\mu_X = E(X) = \sum_{x_i \in R_X} x_i f(x_i) = \int x f(x)dx$$

Properties of Expectation:

1. $E(aX + b) = aE(X) + b$
2. $E(X + Y) = E(X) + E(Y)$
3. $E[g(X)] = \sum_{x \in R_X} g(x)f(x)$ or $\int g(x)f(x)dx$
4. $E(x^2) = \sum x^2 f(x)$

Variance

Variance of X is defined as σ_X^2

$$\sigma_X^2 = V(X) = E\left[(X - \mu_X)^2\right] = E(X^2) - [E(X)]^2$$
$$= \sum_{x \in R_X} (x - \mu_X)^2 f(x) \text{ or } \int_{-\infty}^{\infty} (x - \mu_X)^2 f(x)dx$$

Standard deviation of X is defined as σ_X

$$\sigma_X = \sqrt{V(X)}$$

Properties of Variance:

- $V(aX + b) = a^2 V(X)$

3 Joint Distribution

- **n-dimentional random variable, $(X, Y, ...)$** $= (X(s), Y(s), ...) \forall s \in S$
- **Range Space** $R_{X,Y} = \{(x, y) \mid x = X(s), y = Y(s), \forall s \in S\}$
- **Discrete / Continuous** two-dimensional random variable depends if **both** random variable is countable and finite or vice-versa.

3.1 Joint, Marginal and Conditional Distribution

Joint probability (mass) function, jpf is defined by $(x, y) \in R_{X,Y}$

$$f_{X,Y}(x, y) = P(X = x, Y = y)$$

Well-defined jpf X, Y of both discrete and continuous random variable satisfies:

1. $f_{X,Y}(x, y) \ge 0, \forall (x, y) \in R_{X,Y}$
2. $f_{X,Y}(x, y) = 0, \forall (x, y) \notin R_{X,Y}$
3. $\sum_{i=1}^{\infty}\sum_{j=1}^{\infty} f_{X,Y}(x_i, y_i) = \sum_{i=1}^{\infty}\sum_{j=1}^{\infty} P(X = x_i, Y = y_j) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{X,Y}(x, y)dxdy = 1$

Marginal Probability Distribution

Marginal Probability Distribution of X is defined as sum of $f(y)$ after fixing $X = x$

$$f_X(x) = \sum_y f_{X,Y}(x, y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)dy$$

- $f_X(x)$ satisfies all properties of probability function

Conditional probability Function

Conditional Probability function of Y given $X = x$ is defined as

$$f_{Y \mid X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)}$$

- $f_{Y \mid X}(y \mid x)$ is defined only for x such that $f_X(x) > 0$
- $f_{Y \mid X}(y \mid x)$ is not a probability function and therefore, sum $\neq 1$

Applications:

- $P(Y \le y \mid X = x) = \int_{-\infty}^{y} f_{Y|X}(t|x)dt$
- Regression function $E(Y \mid X = x) = \int_{-\infty}^{\infty} y f_{Y|X}(y|x)dy$

3.2 Independent Random Variables

Random variable $X \perp Y$ are **independent** $\Leftrightarrow$

$$f_{X,Y}(x, y) = f_X(x)f_Y(y)$$
$\Leftrightarrow$ both of the following holds

1. $R_{X,Y}$ spans **Product Space** $R_{X,Y} = \{(x, y) \mid x \in R_X, y \in R_Y\} = R_X \times R_Y$, when probability function is positive.
2. $\forall (x, y) \in R_{X,Y}$

$$f_{X,Y}(x, y) = c \cdot g_1(x) \cdot g_2(y)$$

where g_1 depends only on x, g_2 depends only on y, and c is constant

Let $X \perp Y$

- $P(X \in A; Y \in B) = P(X \in A)P(Y \in B)$
- $P(X \le x; Y \le y) = P(X \le x)P(Y \le y)$
- Let $g1(.), g2(.)$ be arbitrary functions, $g1(X), g2(Y)$ are independent
- $f_X(x) > 0 \Rightarrow f_{Y \mid X}(y \mid x) = f_Y(y)$

3.3 Expectation, Covariance and Correlation Coefficient

Expectation, $E(g(X, Y))$:

$$E(g(X, Y)) = \sum_x \sum_y g(x, y)f_{X,Y}(x, y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x, y)f_{X,Y}(x, y)dydx$$

Covariance, $\text{cov}(X, Y)$

$$\text{cov}(X, Y) = \sum_x \sum_y (x - \mu_x)(y - \mu_y)f_{X,Y}(x, y)$$
$$= E[(X - \mu_X)(Y - \mu_Y)] = E(XY) - \mu_X\mu_Y$$

Properties of covariance:

- $X \perp Y \Rightarrow \text{cov}(X, Y) = 0$, converse is not true
- $X \perp Y \Rightarrow E(XY) = E(X)E(Y)$
- $\text{cov}(aX + b, cY + d) = ac \cdot \text{cov}(X, Y)$
- $\text{cov}(X, Y) = \text{cov}(Y, X)$

Correlation Coefficient, $\rho(X, Y)$

$$\rho(X, Y) = \frac{\text{cov}(X, Y)}{\sigma_X\sigma_Y}, -1 \le \rho(X, Y) \le 1$$

Properties of joint-variance:

- $V(X \pm Y) = V(X) + V(Y) \pm 2\text{ cov}(X, Y)$
- $V(aX + bY) = a^2V(X)b^2V(Y) - 2ab\text{ cov}(X, Y)$
- $V(X_1 + X_2 + ... + X_n) = V(X_1) + V(X_2) + ... + V(X_n) + 2\sum_{j>i} \text{cov}(X_i, X_j)$

4 Probability Distribution

4.1 Discrete Probability Distribution

$q = 1 - p$

Distribution	pmf	cdf	$E(X), V(X)$
Discrete uniform $U(k)$	$f(x) = \frac{1}{k}, x \in R_x$	$F(x) = \frac{\lvert\{x_i \le x\}\rvert}{k}$	$E(X) = \frac{\sum_i x_i}{k}$ $V(X) = \frac{\sum_i x_i^2}{k} - \mu^2$
Bernoulli $X \in \{0, 1\}$	$f(x) = p^x q^{1-x}$	$F(0) = q, F(1) = 1$	$E(X) = p$ $V(X) = pq$
Bin(n, p) # successes	$f(x) = \binom{n}{x}$ $p^x q^{n-x}$	$F(x) = \sum_{i=0}^{x}\binom{n}{i}$ $p^i q^{n-i}$	$E(X) = np$ $V(X) = npq$
NB(k, p) # trials to k successes	$f(x) = \binom{x-1}{k-1}$ $p^k q^{x-k}$	$F(x) = \sum_{i=k}^{x}\binom{i-1}{k-1}$ $p^k q^{i-k}$	$E(X) = \frac{k}{p}$ $V(X) = \frac{kq}{p^2}$
Geom(p) = NB$(1, p)$ # trials to 1st success	$f(x) = q^{x-1}p$	$F(x) = 1 - q^x$	$E(X) = \frac{1}{p}$ $V(X) = \frac{q}{p^2}$
Poisson(λ) # events in an interval	$f(x) = \frac{e^{-\lambda}\lambda^x}{x!}$	$F(x) = \sum_{i=0}^{x}\frac{e^{-\lambda}\lambda^i}{i!}$	$E(X) = \lambda$ $V(X) = \lambda$
Poisson process $\lambda = \alpha T$	same as Poisson	same as Poisson	$E(X) = \alpha T$ $V(X) = \alpha T$

Approximate Binomial using Poisson Distribution

As $n \to \infty$ and $p \to 0$ but $\lambda = np$ remains constant, We may approximate $X \sim \text{Bin}(n, p)$ with $X \sim \text{Poisson}(n, p)$

$$\lim_{p\to 0; n\to\infty} P(X = x) = \frac{e^{-np}(np)^x}{x!}$$

Condition to use:

$$n \ge 20 \wedge p \le 0.05 \text{ or } n \ge 100 \wedge np \le 10$$

4.2 Continuous Probability Distribution

Continuous Uniform Distribution, $X \sim U(a,b)$

$$f_X(x) = \begin{cases} \frac{1}{b-a} a \le x \le b \\ 0 \quad \text{otherwise} \end{cases}, \quad E(x) = \frac{a+b}{2}, \quad V(X) = \frac{(b-a)^2}{12}$$

$$F_X(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \le x \le b \\ 1 & x > b \end{cases}$$

Exponential Distribution, $X \sim \text{Exp}(\lambda)$

X be waiting time until the next event when events occur independently at a constant average rate. For $\lambda > 0$, λ is the average event rate, Mean waiting time is $E(X) = \frac{1}{\lambda}$.

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \ge 0 \\ 0 & \text{if } x < 0 \end{cases}, \quad E(X) = \frac{1}{\lambda}, \quad V(X) = \frac{1}{\lambda^2}$$

$$F_X(x) = P(X \le x) = \begin{cases} 1 - e^{-\lambda x} & \text{if } x \ge 0 \\ 0 & \text{if } x < 0 \end{cases}$$

Hence, for $x \ge 0$,

$$P(X > x) = e^{-\lambda x}$$

Let $\mu = \frac{1}{\lambda}$. Then

$$f_X(x) = \begin{cases} \frac{1}{\mu} e^{-\frac{x}{\mu}} & \text{if } x \ge 0 \\ 0 & \text{if } x < 0 \end{cases}$$

$$E(X) = \mu, \quad V(X) = \mu^2, \quad F_X(x) = 1 - e^{-\frac{x}{\mu}}$$

Normal Distribution, $X \sim N(\mu, \sigma^2)$

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty < x < \infty, E(X) = \mu, V(X) = \sigma^2$$

$$Z = \frac{X - \mu}{\sigma}$$

$$\Phi(z) = F_Z(z) = P(Z < z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

$$P(x_1 < X < x_2) = \Phi\left(\frac{x_2 - \mu}{\sigma}\right) - \Phi\left(\frac{x_1 - \mu}{\sigma}\right)$$

$$P(Z \leftarrow z) = P(Z \ge z) = 1 - \Phi(z)$$

Quartile α

$$P(Z \ge z_\alpha) = \alpha$$

Approximate Binomial using Normal Distribution

As $n \to \infty$ and p is constant, we may approximate binomial distribution using normal distribution.

$$Z = \frac{X - E(X)}{\sqrt{V(X)}} = \frac{X - np}{\sqrt{np(1-p)}}$$

Use when $np > 5$ and $n(1-p) > 5$

Continuity Correction

$$P(X = k) \approx P\left(k - \frac{1}{2} < X < k + \frac{1}{2}\right)$$

$$P(a \le X \le b) \approx P\left(a - \frac{1}{2} < X < b + \frac{1}{2}\right)$$

$$P(a < X \le b) \approx P\left(a + \frac{1}{2} < X < b + \frac{1}{2}\right)$$

$$P(a \le X < b) \approx P\left(a - \frac{1}{2} < X < b - \frac{1}{2}\right)$$

$$P(a < X < b) \approx P\left(a + \frac{1}{2} < X < b - \frac{1}{2}\right)$$

$$P(X \le c) = P(0 \le X \le c) \approx P\left(-\frac{1}{2} < X < c + \frac{1}{2}\right)$$

$$P(X > c) = P(c < X \le n) \approx P\left(c + \frac{1}{2} < X < n + \frac{1}{2}\right)$$

5 Sampling and Sampling Distributions

Sampling is a subset of the population. Population is the total set of all possible observations of an experiment.

- Finite population consist of finite number of observations.
- Infinite population consist of infinitely many (countably or uncountably) observations.

Random Sampling

Simple random sampling (SRS) selects subset n from N population, where every subset has the same probability of being selected.

For a specific subset S of size n, the probability of selecting with replacement is given by, if n is known:

$$P(S) = 1/\binom{N}{n}$$

Statistics and Sampling Distributions

Statistic is a random variable: function that maps a random sample of n observations $(X_1, ..., X_n)$ to a real number.

$$\text{Statistic} : (X_1, ..., X_n) \to f(X_1, ..., X_n)$$

Sampling Distribution is the probability distribution of a statistic. $P\left(\overline{X} = \overline{x}\right)$

5.1 Sample Mean as estimator of Population Mean and Variance

Sample Mean, $\overline{X}$:

$$\overline{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$$

Sample Variance, S^2:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} \left(X_i - \overline{X}\right)^2$$

Sample Mean as estimator of Population Mean and Variance

With multiple random samples, sample means can be used as estimators of population mean and population variance.

Population Mean can be estimated by the expected value of the sample means.

$$\mu_X = E\left(\overline{X}\right) = \mu_{\overline{X}}$$

Population Variance can be estimated by the variance of the sample mean.

$$\frac{\sigma_X^2}{n} = V\left(\overline{X}\right) = \sigma_{\overline{X}}^2$$

- As $n \to \infty, \sigma_{\overline{X}} \to 0 \Rightarrow \overline{X} \to \mu_X$

Standard Error, $\sigma_{\overline{X}}$ is the standard deviation of the sample means.

Law of Large Numbers (LLN)

Law of Large Numbers states that as the sample size $n \to \infty$, the sample mean will converge to the population mean, for any $\varepsilon > 0$.

$$P\left(\left|\overline{X} - \mu_X\right| > \varepsilon\right) \to 0$$

Central Limit Theorem (CLT)

Central Limit Theorem states that as the sample size $n \to \infty$, the distribution of the sample mean will converge to a normal distribution.

$$\text{As } n \to \infty, \overline{X} \to N\left(\mu_X, \frac{\sigma_X^2}{n}\right)$$

$$\frac{\overline{X} - \mu_X}{\sqrt{\sigma_X^2/n}} \to Z \sim N(0,1)$$

Depending on the distribution of the population, different number of samples are needed to approximate the normal distribution.

- For symmetric population with no outliers, $n \in [15, 20]$
- For moderately skewed population (like exponential or χ^2), $n \in [30, 50]$
- For extremely skewed population CLT may not be apppropriate even with 1000 samples.

5.2 Sampling Distributions

Chi-square distribution $\chi^2(n)$ with n degrees of freedom is the distribution of the sum of the squares of n independent standard normal random variables.

$$\chi^2(n) = Z_1^2 + ... + Z_n^2 = \sum_{i=1}^{n} Z_i^2$$

Let $U \sim \chi^2(n)$ with n degrees of freedom,

- $E(U) = n$
- $V(U) = 2n$
- For large n, $U \to N(n, 2n)$

Let $Y_1 \sim \chi^2(n_1)$ and $Y_2 \sim \chi^2(n_2)$ are independent random variables $\Rightarrow Y_1 + Y_2 \sim \chi^2(n_1 + n_2)$

Let $Y \sim \chi^2(n)$, $\chi^2(n;\alpha) \in Y$ and

$$P(Y > \chi^2(n;\alpha)) = \alpha$$

When the **population is normal** $N(\mu, \sigma^2)$, the ratio of sample variance and population variance follows Chi-square distribution with $n-1$ degrees of freedom.

$$(n-1)\frac{S^2}{\sigma^2} = \frac{\sum_{i=1}^{n} (X_i - \mu_X)^2}{\sigma^2} \sim \chi^2(n-1)$$

Student's t-distribution

Student's t-distribution $t(n)$ with n degrees of freedom is ration of two independent random variables

$$t(n) = \frac{Z}{\sqrt{U/n}}; \quad U \sim \chi^2(n), \quad Z \sim N(0,1), \quad Z \perp U$$

Let $T \sim t(n)$,

- $E(T) = 0$
- $V(T) = n/(n-2)$ for $n > 2$
- As $n \to \infty$, $t(n) \to Z$

$$P(T > t_{n;\alpha}) = \alpha$$

When X_i are i.i.d. $N(\mu, \sigma^2)$, the ratio of sample mean and sample standard deviation follows Student's t-distribution with $n-1$ degrees of freedom.

$$\frac{\overline{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$$

F-distribution

F-distribution $F(n_1, n_2)$ with n_1 and n_2 degrees of freedom is the distribution of the ratio of two independent Chi-square random variables with n_1 and n_2 degrees of freedom.

$$F(n_1, n_2) = \frac{Y_1/n_1}{Y_2/n_2}; \quad Y_1 \sim \chi^2(n_1), \quad Y_2 \sim \chi^2(n_2), \quad Y_1 \perp Y_2$$

Let $F \sim F(n_1, n_2)$,

- $E(F) = n_2/(n_2 - 2)$ for $n_2 > 2$
- $V(F) = 2n_2^2(n_1 + n_2 - 2)/\left(n_1(n_2-2)^2(n_2-4)\right)$ for $n_2 > 4$
- For $F \sim F(n_1, n_2)$, $(1/F) \sim F(n_2, n_1)$

$$P\left(F > F_{n_1,n_2;\alpha}\right) = \alpha$$

$$F_{n_1,n_2;1-\alpha} = 1/F_{n_2,n_1;\alpha}$$